ATENA YARAMIREE

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EDUCATION

American University of Sharjah, Sharjah, UAE Jan 2023 – Dec 2025
MS in Biomedical Engineering
Relevant Coursework: Computational Method for Engineering, Human Anatomy and Physiology, Biomechanics Engineering, Drug Delivery, Biomedical Informatics, Biomedical Imaging Technologies, and Biomaterials Engineering.

Islamic Azad University of Tehran Medical Branch, Tehran, Iran Sep 2018 – Feb 2022
BS in Biomedical Engineering (Bioelectric)
Relevant Coursework: Digital Signal Processing, Microprocessors, Ultrasound Devices, Defibrillators, Biomedical Engineering Software, Refractometry and Polarimetry.

PUBLICATIONS

Journal Articles

- **Yaramiri, A.**, Asalh, R. A., Asalh, M. A., AlSawaftah, N., Abuwatfa, W. H., & Husseini, G. A. (2025). A Comprehensive Review of Smart Thermosensitive Nanocarriers for Precision Cancer Therapy. *International Journal of Molecular Sciences*, *26*(15), 7322. https://doi.org/10.3390/ijms26157322
- **Yaramiree A.**, Hannan H., Paul V., Abdelgawad M., Husseini G. A. "Comparative Synthesis of WZB117-Targeted Liposomes by Thin-Film Hydration and Microfluidics for Ultrasound-Triggered Drug Delivery." *In preparation.*

Book Chapters

- **Yaramiree A.**, Ashraf F., Chahrour K., Hatami S., Husseini G. A., Velluto D. "Liposomes in Supramolecular Nanoscience and Drug Delivery Systems." Chapter 4 in *Functionalized Supramolecular Nanotechnology: Advances in Disease Diagnosis and Treatment.* CRC Press, Taylor & Francis Group. *Invited chapter, under review.*
- **Yaramiree A.**, Ashraf F., Chahrour K., Hatami S., Husseini G. A., Velluto D. "Stimuli-Responsive Polymeric Nanoparticles." Chapter 11 in *Functionalized Supramolecular Nanotechnology: Advances in Disease Diagnosis and Treatment.* CRC Press, Taylor & Francis Group. *Invited chapter, under review.*

CONFERENCE PRESENTATIONS

- **Yaramiree A.**, Paul V., Husseini G., Abdelgawad M. "Comparing WZB117-Liposome Synthesis via Microfluidic Methods and Thin-Film Hydration." Poster presentation, *MicroTAS 2026*, Granada, Spain, 18–22 October 2026.

RESEARCH EXPERIENCE

MSc Thesis: Comparing WZB117-Liposome Synthesis via Thin-Film Hydration and Microfluidics
Supervisors: Dr. Ghaleb A. Husseini and Dr. Mohamed Abdelgawad, American University of Sharjah

- Synthesized GLUT1-targeted, WZB117-functionalized liposomes for cancer drug delivery using thin-film hydration and microfluidic methods, and compared the resulting formulations.
- Designed and fabricated microfluidic devices for liposome synthesis, and ran syringe-pump-controlled flow experiments.
- Characterized liposomes using DLS, UV-Vis, FTIR, fluorescence spectroscopy and the Stewart assay.
- Studied ultrasound-triggered drug release and fitted release kinetics to zero-, first- and second-order models in MATLAB and Python.
- Presented results at MicroTAS 2026 (Granada, Spain); manuscript in preparation.

SELECTED PROJECTS:

- **AI & Machine Learning:** Deep learning MRI classification of Alzheimer's disease (ResNet-18, Inception-v3; 94% accuracy); anemia and hemoglobin prediction using regression and classification models.
- **Literature Reviews:** Smart thermosensitive nanocarriers for precision cancer therapy (published, IJMS); smart implantable devices across organ systems; MRI in diagnosing and monitoring neurodegenerative diseases; ECG machine technology, applications and advances; EMG and nerve conduction study devices and test principles (BS).
- **Modelling & Biomechanics:** MATLAB numerical analysis of a gas-phase isothermal packed bed reactor for hydrocarbon dehydrogenation; biomechanics of bone loss in astronauts after space missions.

SKILLS

Laboratory & Research Skills:

- Liposome synthesis (thin-film hydration and microfluidics method)
- Microfluidic device design and fabrication
- Syringe-pump-controlled microfluidic experiments
- Characterization: DLS, FTIR, UV-Vis and fluorescence spectroscopy, Stewart assay
- Drug delivery systems, Nanomaterials and Biomaterials

Technical & Analytical Skills:

- MATLAB, Python
- Data analysis and visualization
- Kinetic modelling and data fitting
- Machine learning and deep learning (healthcare applications)
- Signal processing fundamentals

Tools & Software:

- Microsoft Office, Google Workspace, Mendeley, RefWorks

Languages: English, Arabic, Farsi

WORK EXPERIENCE

Part-Time Assistant Engineering Coordinator | NOI Technologies, Dubai, UAE Jan 2025 – Dec 2025

- Supported engineering operations through documentation and scheduling
- Facilitated communication between engineering teams and stakeholders

Biomedical Engineering Intern, Farhikhtegan Hospital, Tehran, Iran Oct 2021 – Jan 2022

- Responsible for conducting Preventative Maintenance around the hospital; troubleshooting and updating equipment into the database.
- Diagnosed malfunctioning equipment and ordered parts to repair or replace non-working ones.

LEADERSHIP EXPERIENCE

Research Coordinator, IEEE EMBS AUS Student Chapter, Sharjah, UAE Feb 2025 – Dec 2025

- Managed a weekly research content schedule, coordinating a team to produce one to two research articles per week.
- Coordinated semester-long interdisciplinary research projects aimed at journal or conference publication.
- Helped organize the BioPython workshop series and a neuroscience and engineering symposium at AUS.

AWARDS

- Certificate of Completion for Achievement and Participation in the IEEE EMBS AUS Student Chapter and Computer Society BioPython Series 25.
- Certificate of Appreciation: Volunteered as a member of the organization committee for the 10th UAE Graduate Student Research Conference (UAE GSRC25), held on April 12, 2025, at the American University of Sharjah.
- Achieved top 10 ranking in the Microprocessor course, demonstrating strong foundations in embedded systems and low-level computing.
- Ranked 1st and 2nd in the field of fighting robots and water rockets in the 6th Salam Cup Research Competition.